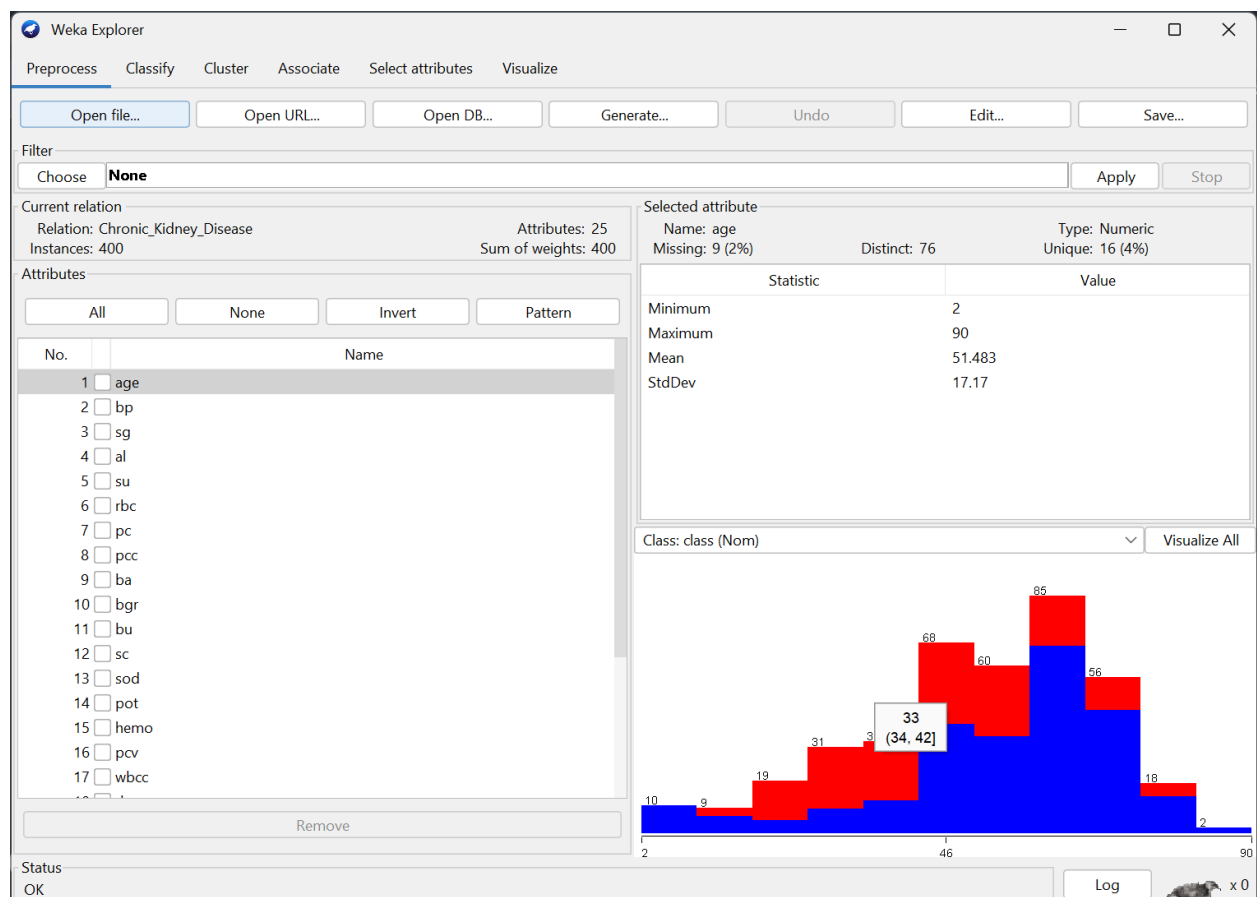


Original data before preprocessing

Relation: Chronic_Kidney_Disease																			
No.	1: age Numeric	2: bp Numeric	3: sg Nominal	4: al Nominal	5: su Nominal	6: rbc Nominal	7: pc Nominal	8: pcc Nominal	9: ba Nominal	10: bgr Numeric	11: bu Numeric	12: sc Numeric	13: sod Numeric	14: pot Numeric	15: hemo Numeric	16: pcv Numeric	17: wbcc Numeric	18: rbcc Numeric	19: htn Nominal
1	48.0	80.0	1.020	1	0		normal	notpr...	notpr...	121.0	36.0	1.2			15.4	44.0	7800.0	5.2	yes
2	7.0	50.0	1.020	4	0		normal	notpr...	notpr...		18.0	0.8			11.3	38.0	6000.0		no
3	62.0	80.0	1.010	2	3	normal	normal	notpr...	notpr...	423.0	53.0	1.8			9.6	31.0	7500.0		no
4	48.0	70.0	1.005	4	0	normal	abnor...	present	notpr...	117.0	56.0	3.8	111.0	2.5	11.2	32.0	6700.0	3.9	yes
5	51.0	80.0	1.010	2	0	normal	normal	notpr...	notpr...	106.0	26.0	1.4			11.6	35.0	7300.0	4.6	no
6	60.0	90.0	1.015	3	0			notpr...	notpr...	74.0	25.0	1.1	142.0	3.2	12.2	39.0	7800.0	4.4	yes
7	68.0	70.0	1.010	0	0		normal	notpr...	notpr...	100.0	54.0	24.0	104.0	4.0	12.4	36.0			no
8	24.0		1.015	2	4	normal	abnor...	notpr...	notpr...	410.0	31.0	1.1			12.4	44.0	6900.0	5.0	no
9	52.0	100.0	1.015	3	0	normal	abnor...	present							10.8	33.0	9600.0	4.0	yes
10	53.0	90.0	1.020	2	0	abnor...	abnor...	present	notpr...	70.0	107.0	7.2	114.0	3.7	9.5	29.0	12100.0	3.7	yes
11	50.0	60.0	1.010	2	4		abnor...	present	notpr...	490.0	55.0	4.0			9.4	28.0			yes
12	63.0	70.0	1.010	3	0		abnor...	notpr...	notpr...	380.0	60.0	2.7	131.0	4.2	10.8	32.0	4500.0	3.8	yes
13	68.0	70.0	1.015	3	1		normal	present	notpr...	208.0	72.0	2.1	138.0	5.8	9.7	28.0	12200.0	3.4	yes
14	68.0	70.0						notpr...	notpr...	98.0	86.0	4.6	135.0	3.4	9.8				yes
15	68.0	80.0	1.010	3	2	normal	abnor...	present	present	157.0	90.0	4.1	130.0	6.4	5.6	16.0	11000.0	2.6	yes
16	40.0	80.0	1.015	3	0		normal	notpr...	notpr...	76.0	162.0	9.6	141.0	4.9	7.6	24.0	3800.0	2.8	yes
17	47.0	70.0	1.015	2	0		normal	notpr...	notpr...	99.0	46.0	2.2	138.0	4.1	12.6				no
18	47.0	80.0						notpr...	notpr...	114.0	87.0	5.2	139.0	3.7	12.1				yes
19	60.0	100.0	1.025	0	3		normal	notpr...	notpr...	263.0	27.0	1.3	135.0	4.3	12.7	37.0	11400.0	4.3	yes
20	62.0	60.0	1.015	1	0		abnor...	present	notpr...	100.0	31.0	1.6			10.3	30.0	5300.0	3.7	yes
21	61.0	80.0	1.015	2	0	abnor...	abnor...	notpr...	notpr...	173.0	148.0	3.9	135.0	5.2	7.7	24.0	9200.0	3.2	yes
22	60.0	90.0						notpr...	notpr...			180.0	4.5		10.9	32.0	6200.0	3.6	yes
23	48.0	80.0	1.025	4	0	normal	abnor...	notpr...	notpr...	95.0	163.0	7.7	136.0	3.8	9.8	32.0	6900.0	3.4	yes
24	31.0	70.0	1.010	0	0		normal	notpr...	notpr...										

Relation: Chronic_Kidney_Disease																			
No.	1: age Numeric	2: bp Numeric	3: sg Nominal	4: al Nominal	5: su Nominal	6: rbc Nominal	7: pc Nominal	8: pcc Nominal	9: ba Nominal	10: bgr Numeric	11: bu Numeric	12: sc Numeric	13: sod Numeric	14: pot Numeric	15: hemo Numeric	16: pcv Numeric	17: wbcc Numeric	18: rbcc Numeric	19: htn Nominal
70	26.0	70.0	1.015	0	4		normal	notpr...	notpr...	250.0	20.0	1.1			15.6	52.0	6900.0	6.0	no
71	61.0	80.0	1.015	0	4		normal	notpr...	notpr...	360.0	19.0	0.7	137.0	4.4	15.2	44.0	8300.0	5.2	yes
72	46.0	60.0	1.010	1	0	normal	normal	notpr...	notpr...	163.0	92.0	3.3	141.0	4.0	9.8	28.0	14600.0	3.2	yes
73	64.0	90.0	1.010	3	3		abnor...	present	notpr...		35.0	1.3			10.3				yes
74		100.0	1.015	2	0	abnor...	abnor...	notpr...	notpr...	129.0	107.0	6.7	132.0	4.4	4.8	14.0	6300.0		yes
75	56.0	90.0	1.015	2	0						107.0	6.7	131.0	4.8	9.1	29.0	6400.0	3.4	yes
76	5.0		1.015	1	0		normal	notpr...	notpr...		16.0	0.7	138.0	3.2	8.1				no
77	48.0	80.0	1.005	4	0	abnor...	abnor...	present	present	133.0	139.0	8.5	132.0	5.5	10.3	36.0	6200.0	4.0	no
78	67.0	70.0	1.010	1	0		normal	notpr...	notpr...	102.0	48.0	3.2	137.0	5.0	11.9	34.0	7100.0	3.7	yes
79	70.0	80.0						notpr...	notpr...	158.0	85.0	3.2	141.0	3.5	10.1	30.0			yes
80	56.0	80.0	1.010	1	0		normal	notpr...	notpr...	165.0	55.0	1.8			13.5	40.0	11800.0	5.0	yes
81	74.0	80.0	1.010	0	0		normal	notpr...	notpr...	132.0	98.0	2.8	133.0	5.0	10.8	31.0	9400.0	3.8	yes
82	45.0	90.0						notpr...	notpr...	360.0	45.0	2.4	128.0	4.4	8.3	29.0	5500.0	3.7	yes
83	38.0	70.0						notpr...	notpr...	104.0	77.0	1.9	140.0	3.9					yes
84	48.0	70.0	1.015	1	0	normal	normal	notpr...	notpr...	127.0	19.0	1.0	134.0	3.6					yes
85	59.0	70.0	1.010	3	0	normal	abnor...	notpr...	notpr...	76.0	186.0	15.0	135.0	7.6	7.1	22.0	3800.0	2.1	yes
86	70.0	70.0	1.015	2				notpr...	notpr...		46.0	1.5			9.9				no
87	56.0	80.0						notpr...	notpr...	415.0	37.0	1.9							no
88	70.0	100.0	1.005	1	0	normal	abnor...	present	notpr...	169.0	47.0	2.9			11.1	32.0	5800.0	5.0	yes
89	58.0	110.0	1.010	4	0		normal	notpr...	notpr...	251.0	52.0	2.2					13200.0	4.7	yes
90	50.0	70.0	1.020	0	0		normal	notpr...	notpr...	109.0	32.0	1.4	139.0	4.7					no
91	63.0	100.0	1.010	2	2	normal	normal	notpr...	present	280.0	35.0	3.2	143.0	3.5	13.0	40.0	9800.0	4.2	yes
92	56.0	70.0	1.015	4	1	abnor...	normal	notpr...	notpr...	210.0	26.0	1.7	136.0	3.8	16.1	52.0	12500.0	5.6	no
93	71.0	70.0	1.010	3	0	normal	abnor...	present	present	219.0	82.0	2.6	133.0	4.4	10.4	33.0	5600.0	3.6	yes



Problem is that we have missing values.

FIX.

Remove the missing values

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

Filter

Numeric to Nominal

Numeric Transform

Obfuscate

Ordinal to Numeric

Partitioned Multi Filter

PKIDiscretize

Principal Components

Random Projection

Random Subset

Remove

Remove By Name

Remove Type

Remove Useless

Rename Attribute

Rename Nominal Values

Reorder

Replace Missing Values

Replace Missing With User Constant

Replace

Sort Labels

Standardize

String to Date

String to Integer

Swap Values

Time Series

Time Series

Transpose

instance

Attributes: 25

Sum of weights: 400

Pattern

Selected attribute

Name: age

Missing: 0 (0%)

Distinct: 77

Type: Numeric

Unique: 16 (4%)

Statistic	Value
Minimum	2
Maximum	90
Mean	51.483
StdDev	16.975

Class: class (Nom)

Visualize All

Replaces all missing values for nominal and numeric attributes in a dataset with the modes and means from the training data

The class attribute is skipped by default.

CAPABILITIES

Class -- Binary class, Date class, Empty nominal class, Missing class values, No class, Nominal class, Numeric class, Relational class, String class, Unary class

Attributes -- Binary attributes, Date attributes, Empty nominal attributes, Missing values, Nominal attributes, Numeric attributes, Relational attributes, String attributes

Interfaces -- Sourcable, UnsupervisedFilter, WeightedAttributesHandler, WeightedInstancesHandler

Additional

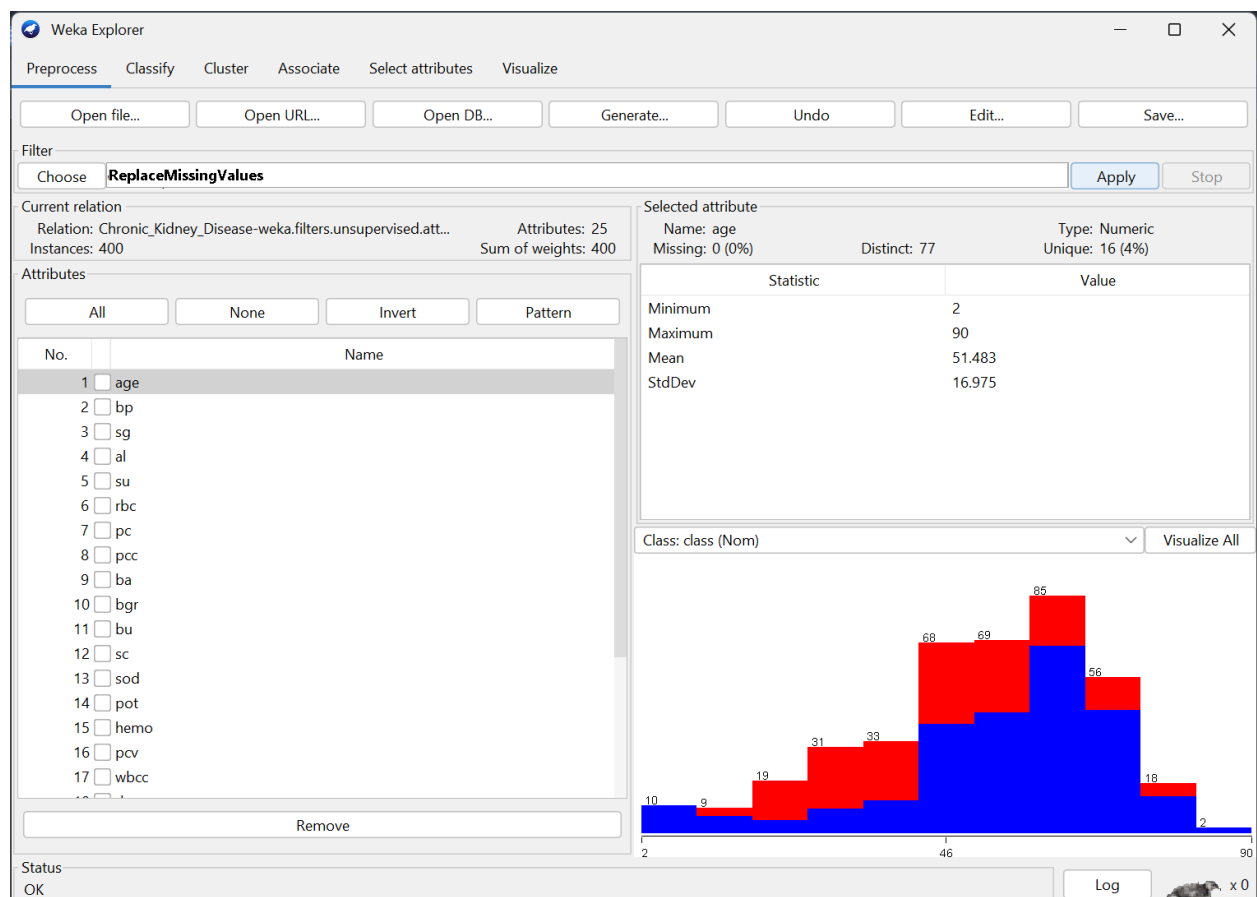
Minimum number of instances: 0

Status

OK

Log

x 0



Used the replacemissingvalue attribute in the choose tab during preprocessing stage.
Missing in this case is 0%.

NO more missing values.

Viewer

Relation: Chronic_Kidney_Disease-weka.filters.unsupervised.attribute.ReplaceMissingValues

No.	1: age Numeric	2: bp Numeric	3: sg Nominal	4: al Nominal	5: su Nominal	6: rbc Nominal	7: pc Nominal	8: pcc Nominal	9: ba Nominal	10: bgr Numeric	11: bu Numeric	12: sc Numeric	13: sod Numeric	14: pot Numeric	15: hemo Numeric	16: pcv Numeric	17: wbcc Numeric	18: rbcc Numeric	19: htn Nominal
1	48.0	80.0	1.020	1	0	normal	normal	notpr...	notpr...	121.0	36.0	1.2	137.5...	4.627...	15.4	44.0	7800.0	5.2	yes
2	7.0	50.0	1.020	4	0	normal	normal	notpr...	notpr...	148.0...	18.0	0.8	137.5...	4.627...	11.3	38.0	6000.0	4.7074...	no
3	62.0	80.0	1.010	2	3	normal	normal	notpr...	notpr...	423.0	53.0	1.8	137.5...	4.627...	9.6	31.0	7500.0	4.7074...	no
4	48.0	70.0	1.005	4	0	normal	abnor...	present	notpr...	117.0	56.0	3.8	111.0	2.5	11.2	32.0	6700.0	3.9	yes
5	51.0	80.0	1.010	2	0	normal	normal	notpr...	notpr...	106.0	26.0	1.4	137.5...	4.627...	11.6	35.0	7300.0	4.6	no
6	60.0	90.0	1.015	3	0	normal	normal	notpr...	notpr...	74.0	25.0	1.1	142.0	3.2	12.2	39.0	7800.0	4.4	yes
7	68.0	70.0	1.010	0	0	normal	normal	notpr...	notpr...	100.0	54.0	24.0	104.0	4.0	12.4	36.0	8406.1...	4.7074...	no
8	24.0	76.46...	1.015	2	4	normal	abnor...	notpr...	notpr...	410.0	31.0	1.1	137.5...	4.627...	12.4	44.0	6900.0	5.0	no
9	52.0	100.0	1.015	3	0	normal	abnor...	present	notpr...	138.0	60.0	1.9	137.5...	4.627...	10.8	33.0	9600.0	4.0	yes
10	53.0	90.0	1.020	2	0	abnor...	abnor...	present	notpr...	70.0	107.0	7.2	114.0	3.7	9.5	29.0	12100.0	3.7	yes
11	50.0	60.0	1.010	2	4	normal	abnor...	present	notpr...	490.0	55.0	4.0	137.5...	4.627...	9.4	28.0	8406.1...	4.7074...	yes
12	63.0	70.0	1.010	3	0	abnor...	abnor...	present	notpr...	380.0	60.0	2.7	131.0	4.2	10.8	32.0	4500.0	3.8	yes
13	68.0	70.0	1.015	3	1	normal	normal	present	notpr...	208.0	72.0	2.1	138.0	5.8	9.7	28.0	12200.0	3.4	yes
14	68.0	70.0	1.020	0	0	normal	normal	notpr...	notpr...	98.0	86.0	4.6	135.0	3.4	9.8	38.88...	8406.1...	4.7074...	yes
15	68.0	80.0	1.010	3	2	normal	abnor...	present	present	157.0	90.0	4.1	130.0	6.4	5.6	16.0	11000.0	2.6	yes
16	40.0	80.0	1.015	3	0	normal	normal	notpr...	notpr...	76.0	162.0	9.6	141.0	4.9	7.6	24.0	3800.0	2.8	yes
17	47.0	70.0	1.015	2	0	normal	normal	notpr...	notpr...	99.0	46.0	2.2	138.0	4.1	12.6	38.88...	8406.1...	4.7074...	no
18	47.0	80.0	1.020	0	0	normal	normal	notpr...	notpr...	114.0	87.0	5.2	139.0	3.7	12.1	38.88...	8406.1...	4.7074...	yes
19	60.0	100.0	1.025	0	3	normal	normal	notpr...	notpr...	263.0	27.0	1.3	135.0	4.3	12.7	37.0	11400.0	4.3	yes
20	62.0	60.0	1.015	1	0	normal	abnor...	present	notpr...	100.0	31.0	1.6	137.5...	4.627...	10.3	30.0	5300.0	3.7	yes
21	61.0	80.0	1.015	2	0	abnor...	abnor...	notpr...	notpr...	173.0	148.0	3.9	135.0	5.2	7.7	24.0	9200.0	3.2	yes
22	60.0	90.0	1.020	0	0	normal	normal	notpr...	notpr...	148.0...	180.0	76.0	4.5	4.627...	10.9	32.0	6200.0	3.6	yes
23	48.0	80.0	1.025	4	0	normal	abnor...	notpr...	notpr...	95.0	163.0	7.7	136.0	3.8	9.8	32.0	6900.0	3.4	yes
24	31.0	70.0	1.010	0	0	normal	normal	notpr...	notpr...	148.0...	57.43...	3.073...	137.5...	4.627...	13.564...	38.88...	8406.1...	4.7074...	no

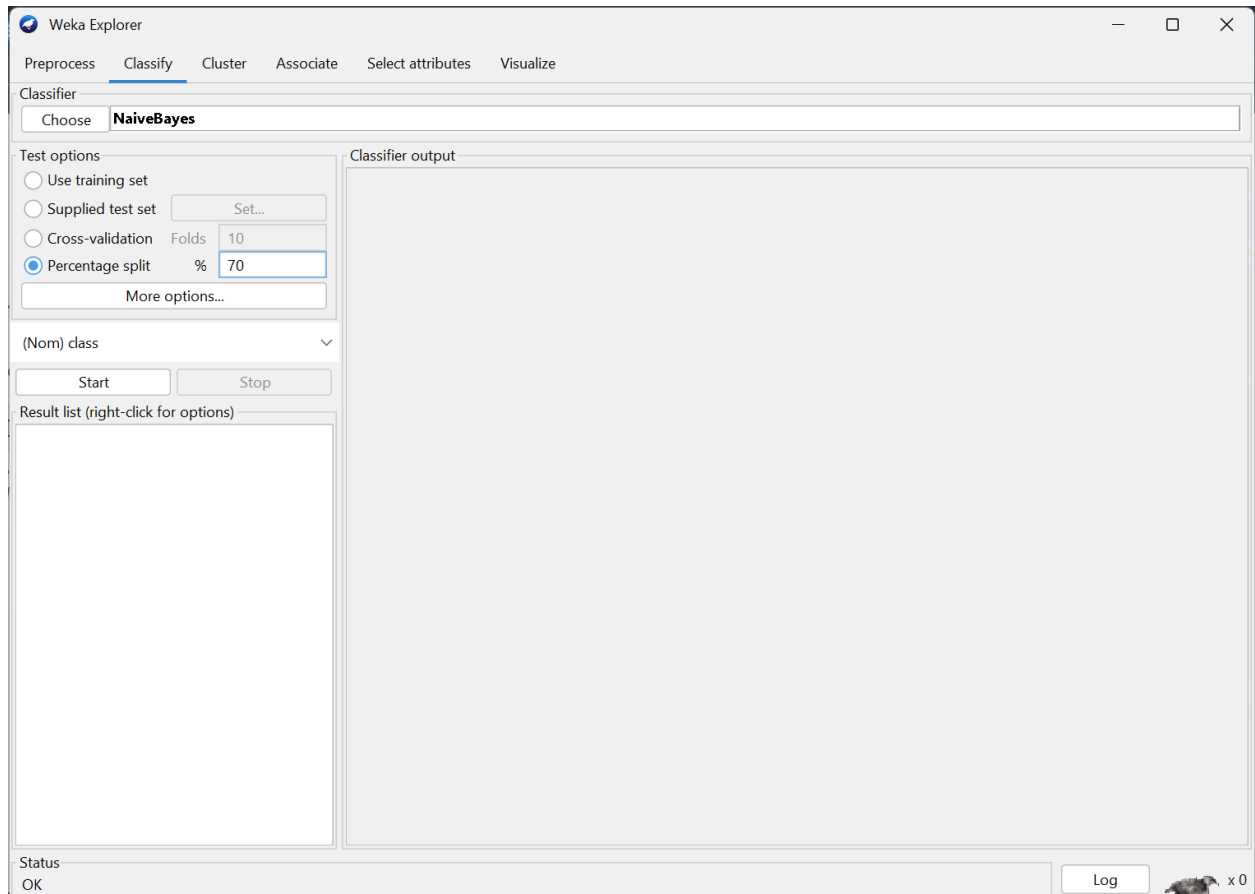
Add instance

Undo

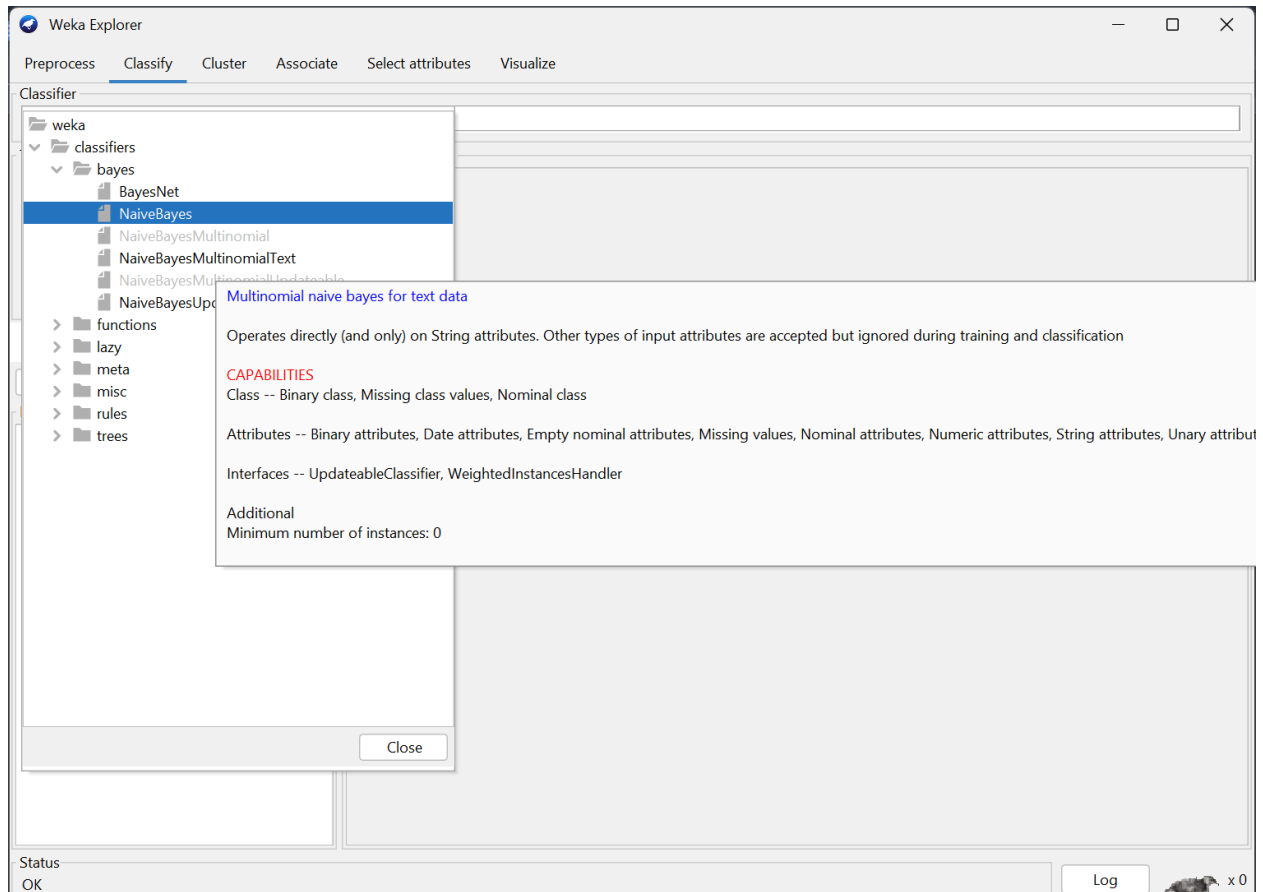
OK

Cancel

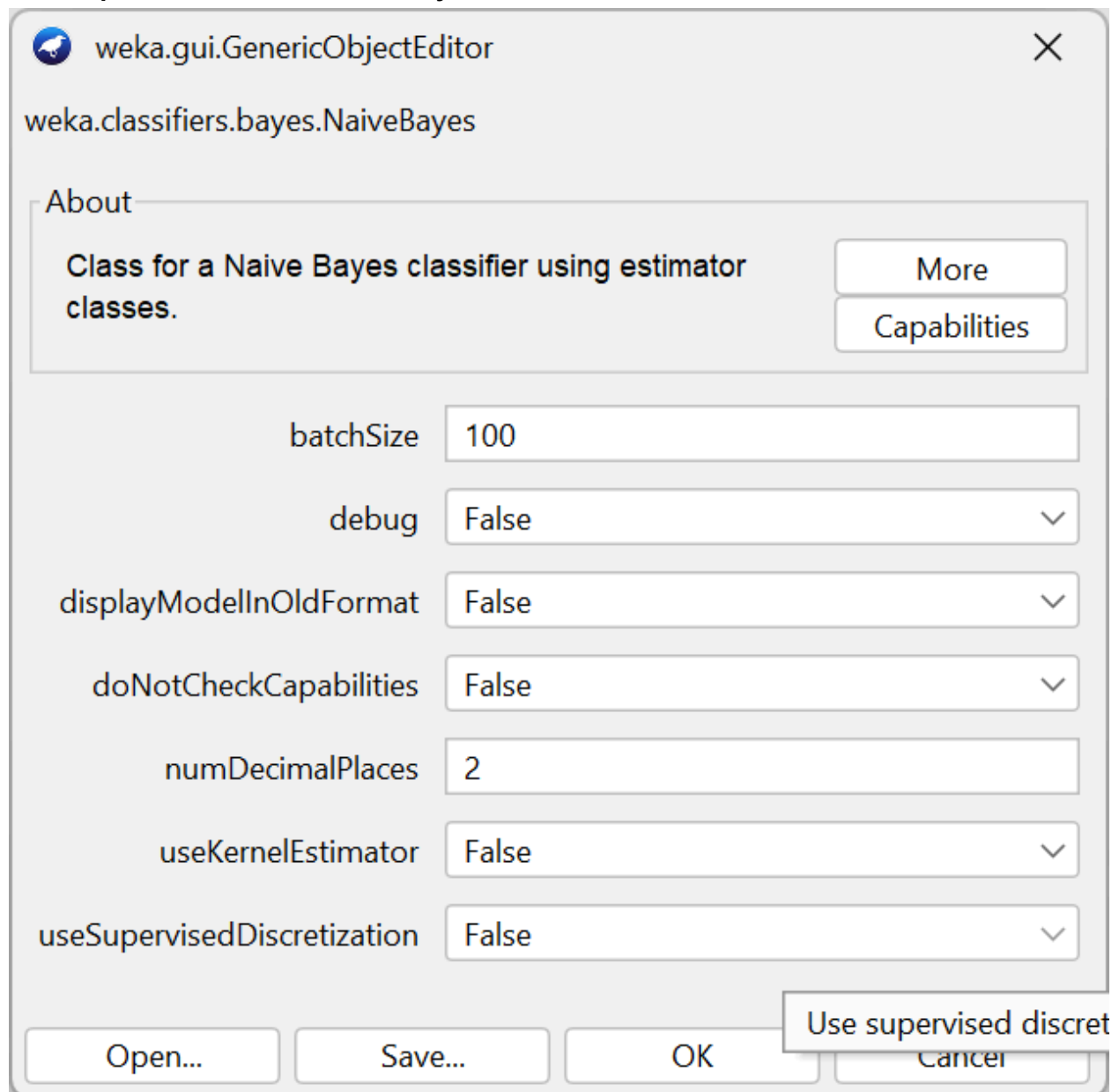
Setting up the test split



Selecting Naive Bayes



Default parameters set for naive bayes



The image shows a screenshot of the Weka GUI's GenericObjectEditor window for the NaiveBayes classifier. The window title is 'weka.gui.GenericObjectEditor' and the path is 'weka.classifiers.bayes.NaiveBayes'. It features an 'About' section with a description and buttons for 'More' and 'Capabilities'. Below this are several parameter settings, each with a label and a text or dropdown field. At the bottom are buttons for 'Open...', 'Save...', 'OK', and 'Cancel'. A tooltip is visible over the 'OK' button.

weka.gui.GenericObjectEditor

weka.classifiers.bayes.NaiveBayes

About

Class for a Naive Bayes classifier using estimator classes.

More

Capabilities

batchSize 100

debug False

displayModelInOldFormat False

doNotCheckCapabilities False

numDecimalPlaces 2

useKernelEstimator False

useSupervisedDiscretization False

Open... Save... OK Cancel

Use supervised discret

Overall output

Classifier output

```
Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances      115          95.8333 %
Incorrectly Classified Instances     5           4.1667 %
Kappa statistic                     0.9124
Mean absolute error                  0.0481
Root mean squared error              0.2094
Relative absolute error              10.2845 %
Root relative squared error          43.4435 %
Total Number of Instances           120

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                0.934    0.000    1.000     0.934    0.966      0.916    1.000    1.000    ckd
                1.000    0.066    0.898     1.000    0.946      0.916    1.000    0.999    notckd
Weighted Avg.   0.958    0.024    0.963     0.958    0.959      0.916    1.000    1.000

=== Confusion Matrix ===

  a  b  <-- classified as
71  5  |  a = ckd
 0 44  |  b = notckd
```

Accuracy : 95.833

Confusion matrix

```
=== Confusion Matrix ===

  a  b  <-- classified as
71  5  |  a = ckd
 0 44  |  b = notckd
```

Analysis:

We had a test set of 120 patients(30%) :

71 With CKD were correctly classified
5 with CKD were classified as healthy
44 healthy were classified correctly(notCKD)
0 healthy patients were classified to have CKD

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) * 100\%$$

$$(71 + 44) / (71 + 44 + 0 + 5) * 100\%$$

95.833%

$$\text{Precision} = TP / (TP + FP) * 100$$

$$71 / (71 + 0) * 100 = 100\%$$

$$\begin{aligned}\text{Sensitivity} &= TP / (TP + FN) * 100 \\ &= 71 / (71 + 5) * 100 \\ &= 93.42\%\end{aligned}$$

$$\begin{aligned}\text{Specificity} &= TN / (TN + FP) \\ &= 44 / (44 + 0) * 100 \\ &= 100\%\end{aligned}$$

Why Naive Bayes performs well here?

Naive Bayes works on the assumption that all attributes are **conditionally independent** given the class. The CKD dataset has many attributes (hemoglobin, serum creatinine, blood urea, etc.) that each independently carry strong diagnostic signal, which suits this assumption well — hence the high accuracy.

Correlation Check among the attributes **CorrelationAttributeEval**

Weka Explorer

Preprocess Classify Cluster Associate **Select attributes** Visualize

Attribute Evaluator
Choose **CorrelationAttributeEval**

Search Method
Choose **Ranker** -T -1.7976931348623157E308 -N -1

Attribute Selection Mode
☒ Use full training set
☐ Cross-validation Folds 10 Seed 1

(Nom) class
Start Stop

Result list (right-click for options)
23:17:33 - Ranker + CorrelationAttributeEval

Attribute selection output

Correlation Ranking Filter

Ranked attributes:

0.7296	15	hemo
0.6901	16	pcv
0.5909	18	rbcc
0.5904	19	htn
0.5591	20	dm
0.477	4	al
0.4014	10	bgr
0.3933	22	appet
0.3752	7	pc
0.3752	23	pe
0.372	11	bu
0.3505	3	sg
0.3423	13	sod
0.3254	24	ane
0.3009	5	su
0.2941	12	sc
0.2906	2	bp
0.2826	6	rbc
0.2653	8	pcc
0.2361	21	cad
0.2254	1	age
0.2053	17	wbcc
0.1869	9	ba
0.0769	14	pot

Selected attributes: 15,16,18,19,20,4,10,22,7,23,11,3,13,24,5,12,2,6,8,21,1,17,9,14 : 24

Status
OK

Log x 0

InfoGainAttributeEval

Weka Explorer

Preprocess Classify Cluster Associate **Select attributes** Visualize

Attribute Evaluator
Choose **InfoGainAttributeEval**

Search Method
Choose **Ranker** -T -1.7976931348623157E308 -N -1

Attribute Selection Mode
☒ Use full training set
☐ Cross-validation Folds Seed

(Nom) class

Result list (right-click for options)
23:17:33 - Ranker + CorrelationAttributeEval
23:19:24 - Ranker + InfoGainAttributeEval

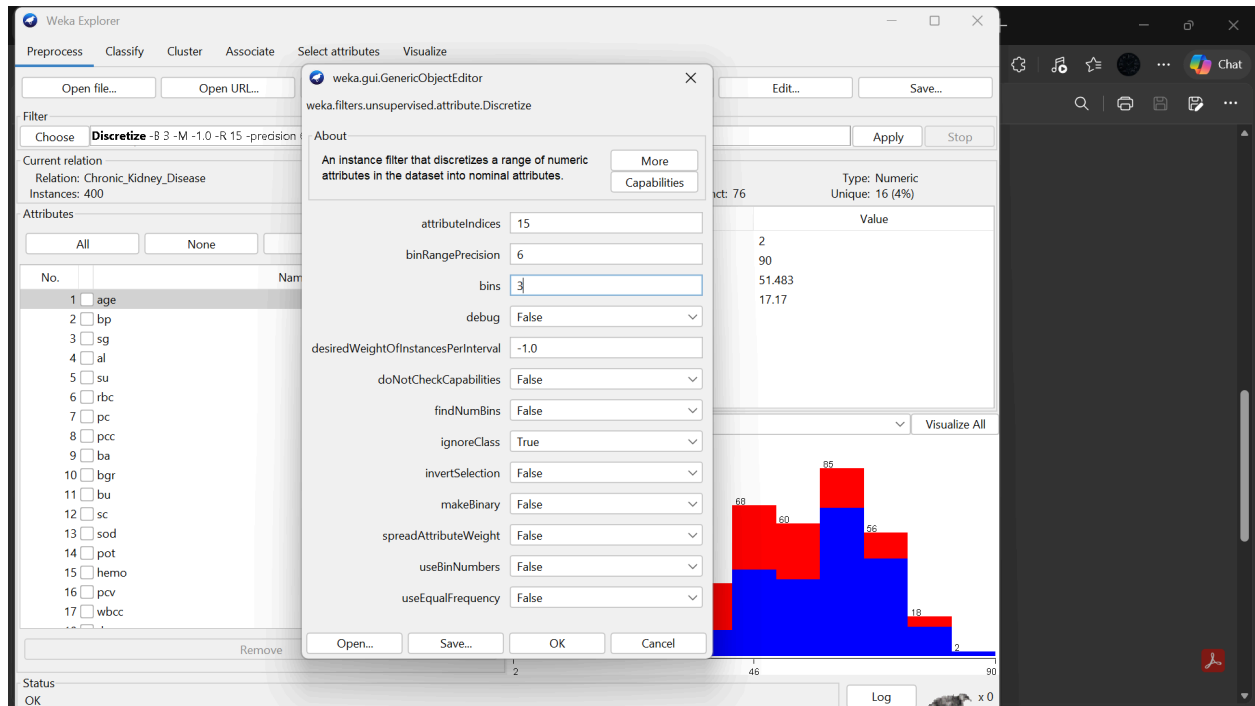
Attribute selection output
Attribute Evaluator (Supervised, class (nominal), 25 class):
Information Gain Ranking Filter

Ranked attributes:
0.6523 15 hemo
0.6079 16 pcv
0.5634 18 rbcc
0.4941 12 sc
0.4565 3 sg
0.3644 4 al
0.3428 13 sod
0.3378 19 htn
0.3318 14 pot
0.3064 20 dm
0.291 11 bu
0.2826 10 bgr
0.237 17 wbcc
0.1659 2 bp
0.1613 22 appet
0.1476 7 pc
0.1476 23 pe
0.115 5 su
0.1129 24 ane
0.1101 1 age
0.0863 6 rbc
0.0765 8 pcc
0.061 21 cad
0.0387 9 ba

Selected attributes: 15,16,18,12,3,4,13,19,14,20,11,10,17,2,22,7,23,5,24,1,6,8,21,9 : 24

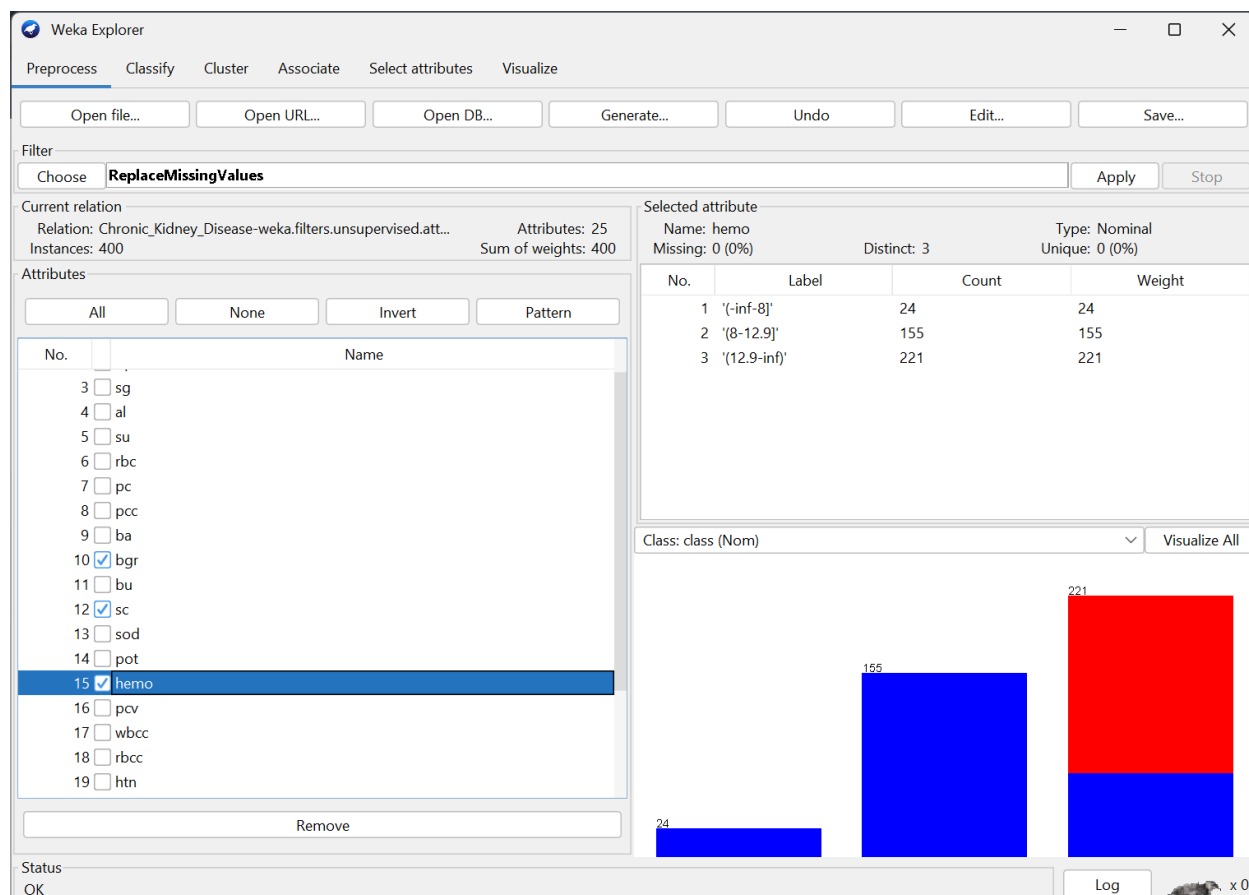
Status
OK x 0

5.Discretization (



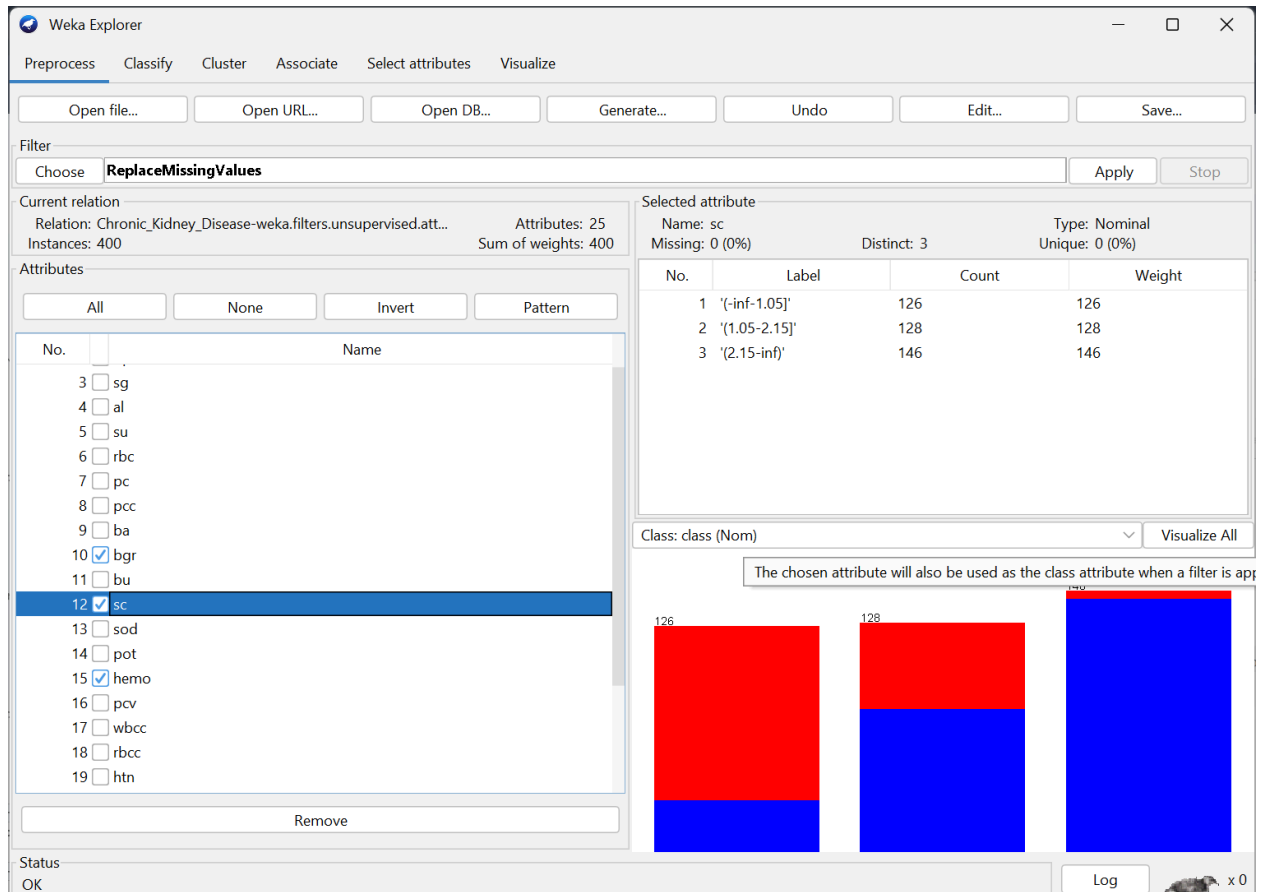
So we first convert the **hemoglobin(hemo)** first into **3 different bins**

After discretization we obtain the following:



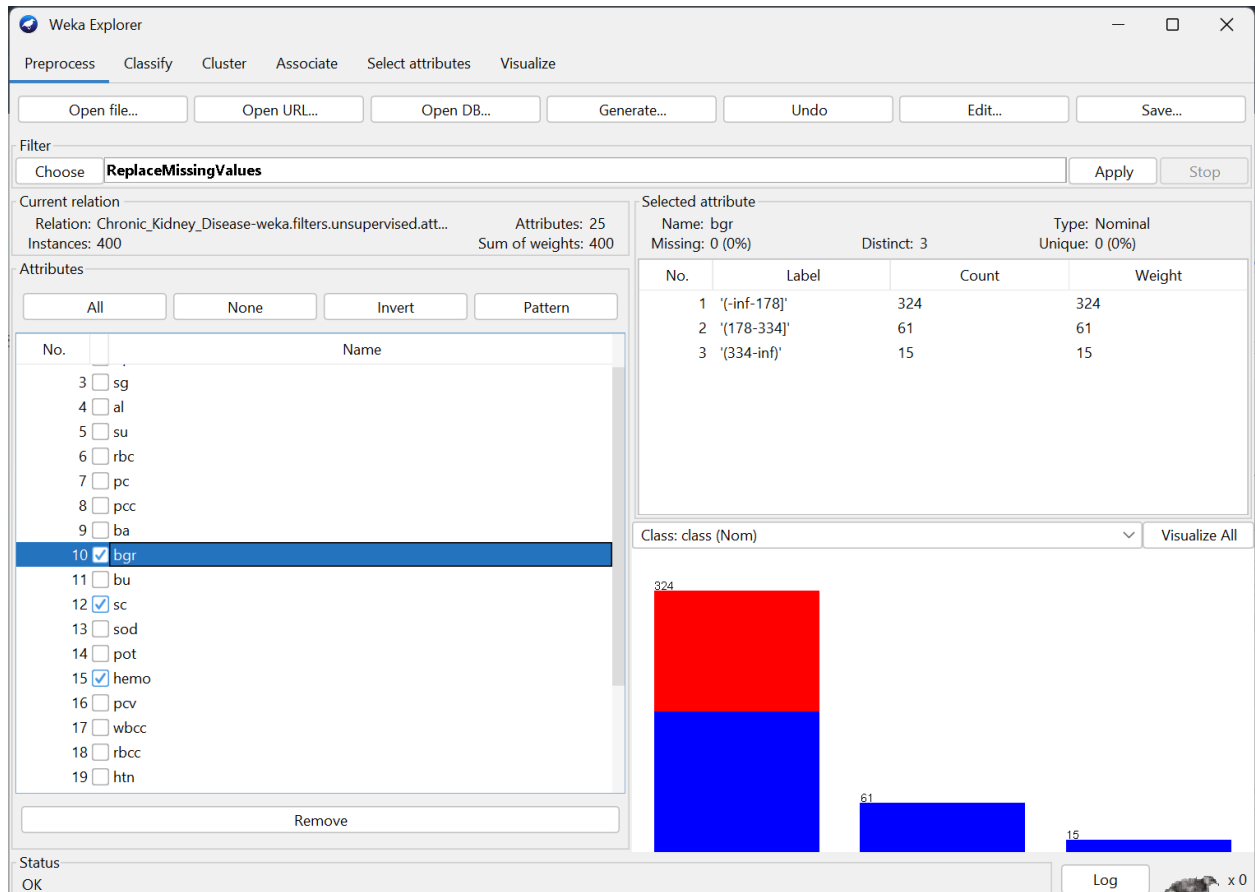
<u>Bin</u>	<u>WEKA Label</u>	<u>Count</u>	<u>Medical Meaning</u>
1	(-inf-8]	24 patients	Severely Low — Severe Anemia
2	(8-12.9]	155 patients	Low-Normal — Mild/Moderate Anemia
3	(12.9-inf)	221 patients	Normal-High — Healthy range

Repeat the same steps for Serum Creatinine(sc):



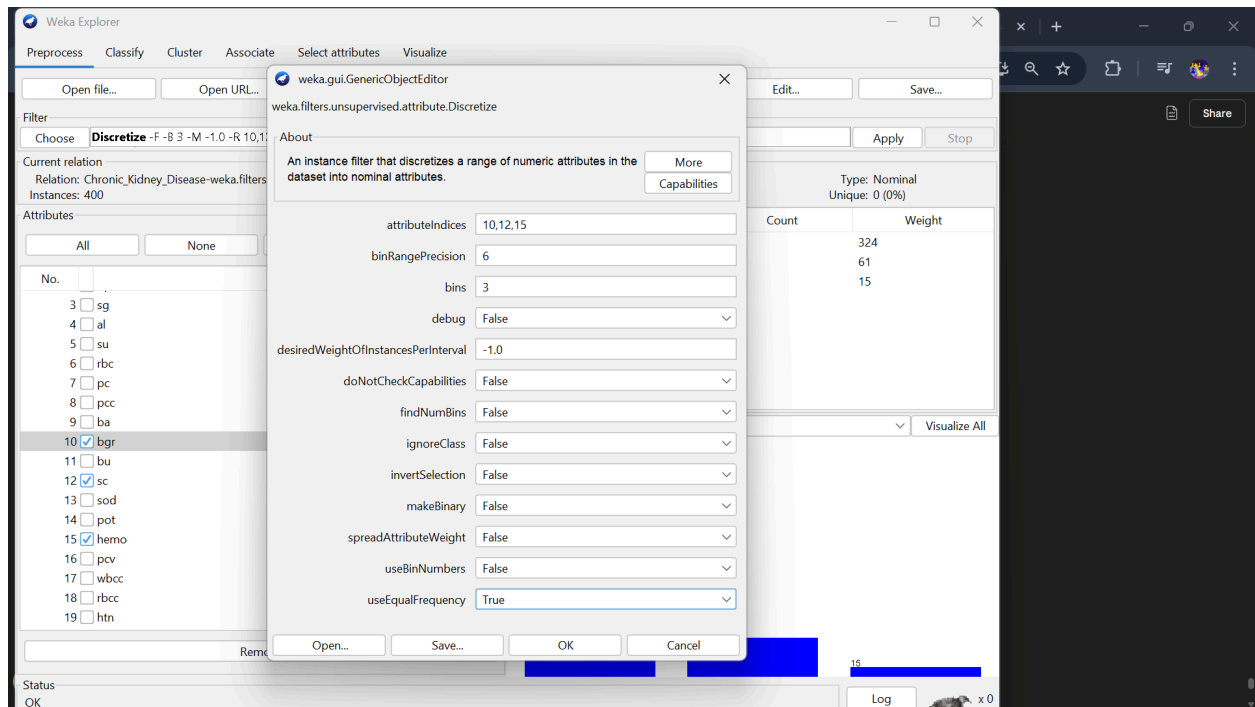
Bin	WEKA Label	Count	Medical Meaning
1	(-inf-1.05]	126 patients	Normal — Healthy kidney function
2	(1.05-2.15]	128 patients	Mildly High — Early kidney damage
3	(2.15-inf)	146 patients	High — Significant kidney failure

Blood Glucose(bgr):



<u>Bin</u>	<u>WEKA Label</u>	<u>Count</u>	<u>Medical Meaning</u>
1	(-inf-178]	324 patients	Normal to Pre-Diabetic
2	(178-334]	61 patients	Diabetic Range
3	(334-inf)	15 patients	Severely Diabetic

Final step



All of our three attributes are now nominal for form numeric form and we can now use them for our algorithms and not any changes we may obtain.

New Output from each of the initial algorithms used: !!!!start from here

Naive Bayes:

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options:

- ☐ Use training set
- ☐ Supplied test set (Set...)
- ☐ Cross-validation (Folds: 10)
- ☒ Percentage split (%: 70)

More options...

(Nom) class: Start Stop

Result list (right-click for options): 22:03:21 - bayes.NaiveBayes

Classifier output:

```

Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances      114      95 %
Incorrectly Classified Instances     6       5 %
Kappa statistic                    0.8953
Mean absolute error                 0.0448
Root mean squared error             0.1996
Relative absolute error             9.5756 %
Root relative squared error        41.4077 %
Total Number of Instances          120

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC  ROC Area  PRC Area  Class
0.921    0.000    1.000    0.921    0.959    0.900    1.000    1.000    ckd
1.000    0.079    0.880    1.000    0.936    0.900    1.000    1.000    notckd
Weighted Avg.  0.950    0.029    0.956    0.950    0.951    0.900    1.000    1.000

=== Confusion Matrix ===

 a  b  <-- classified as
70  6  |  a = ckd
 0 44 |  b = notckd

```

Status: OK Log x 0

From the confusion matrix:
Comparison Table

Metric	Before	After	Change
Accuracy	95.83%	95.00%	-0.83%
Precision	100%	100%	No change
Recall	93.42%	92.10%	-1.32%
Specificity	100%	100%	No change

Observation:

Converting the three attributes into nominal slightly reduces the accuracy of naive bayes algorithm and also the recall while the precision and specificity remain the unaltered

KNN algorithm

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier is 'IBk -K 1 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A "weka.core.EuclideanDistance -R first-last"'. The test options are set to 'Percentage split' at 70%. The classifier output shows the following results:

```
Time taken to build model: 0 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.03 seconds

=== Summary ===

Correctly Classified Instances      118      98.3333 %
Incorrectly Classified Instances     2      1.6667 %
Kappa statistic                    0.9645
Mean absolute error                 0.0201
Root mean squared error             0.1287
Relative absolute error             4.2951 %
Root relative squared error        26.6956 %
Total Number of Instances         120

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      0.974    0.000    1.000     0.974    0.987     0.965    0.987     0.990    ckd
      1.000    0.026    0.957     1.000    0.978     0.965    0.987     0.957    notckd
Weighted Avg.   0.983    0.010    0.984     0.983    0.983     0.965    0.987     0.978

=== Confusion Matrix ===

  a  b  <-- classified as
74  2  |  a = ckd
 0 44 |  b = notckd
```

Comparison Table:

Metric	Before	After	change
Accuracy	95.83%	98.33%	-2.50%
Precision	100%	100%	No change
Recall	93.42%	97.37%	-3.95%
Specificity	100%	100%	No change

Observation:

Converting the three attributes into nominal slightly reduces the accuracy of KNN algorithm and also the recall while the precision and specificity remain the unaltered

Random Forest:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'RandomForest' with parameters '-P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1'. The 'Test options' section shows 'Percentage split' selected with a value of 70%. The 'Result list' on the left shows three entries: '22:03:21 - bayes.NaiveBayes', '22:13:02 - lazy.IBk', and '22:14:11 - trees.RandomForest' (highlighted). The 'Classifier output' pane displays the following results:

```
Time taken to build model: 0.13 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances      120          100 %
Incorrectly Classified Instances     0           0 %
Kappa statistic                     1
Mean absolute error                  0.0482
Root mean squared error              0.092
Relative absolute error              10.3046 %
Root relative squared error          19.0899 %
Total Number of Instances           120

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
      1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    ckd
      1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000    notckd
Weighted Avg.   1.000    0.000    1.000     1.000    1.000     1.000    1.000    1.000

=== Confusion Matrix ===

 a b  <-- classified as
76  0 | a = ckd
 0 44 | b = notckd
```

The status bar at the bottom shows 'Status OK' and a 'Log' button.

Comparison Table:

Observation

Random Forest proved to be a highly effective classifier for the CKD dataset. It performed best with continuous numeric attributes and a reduced clean feature set. Discretization slightly reduced performance showing that preserving numeric precision is important for this algorithm and this dataset.

Metric	before	after	Discretized
Accuracy	100%	100%	98.33%
Precision	100%	100%	98.4%
Recall	100%	100%	98.3%
Specificity	100%	100%	97.4%

Neural Network (MLP)

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **MultilayerPerceptron** -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H a

Test options

☐ Use training set

☐ Supplied test set

☐ Cross-validation Folds

☒ Percentage split %

(Nom) class

Result list (right-click for options)

- 22:03:21 - bayes.NaiveBayes
- 22:13:02 - lazy.IBk
- 22:14:11 - trees.RandomForest
- 22:16:28 - functions.MultilayerPerceptron**

Classifier output

Time taken to build model: 2.58 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances	120	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.0048		
Root mean squared error	0.0318		
Relative absolute error	1.0297	%	
Root relative squared error	6.5945	%	
Total Number of Instances	120		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	ckd
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	notckd
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

```

a b  <-- classified as
76 0 | a = ckd
0 44 | b = notckd

```

Status OK

x 0

Comparison Table:

Decision Tree

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☐ Cross-validation (Folds: 10)
☒ Percentage split (%: 70)
 More options...

(Nom) class: **Start** **Stop**

Result list (right-click for options):
 22:03:21 - bayes.NaiveBayes
 22:13:02 - lazy.IBk
 22:14:11 - trees.RandomForest
 22:16:28 - functions.MultilayerPerceptron
22:20:36 - trees.J48

Classifier output

Time taken to build model: 0.04 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances	113	94.1667 %
Incorrectly Classified Instances	7	5.8333 %
Kappa statistic	0.8713	
Mean absolute error	0.0818	
Root mean squared error	0.2357	
Relative absolute error	17.4864 %	
Root relative squared error	48.9042 %	
Total Number of Instances	120	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.987	0.136	0.926	0.987	0.955	0.875	0.927	0.927	ckd
	0.864	0.013	0.974	0.864	0.916	0.875	0.927	0.882	notckd
Weighted Avg.	0.942	0.091	0.944	0.942	0.941	0.875	0.927	0.911	

=== Confusion Matrix ===

```

a b  <-- classified as
75  1 | a = ckd
 6 38 | b = notckd

```

Status: OK

Log x 0

Comparison Table:

Converting two attributes into nominal does not change the accuracy , precision, recall and the specificity of the decision tree. This might have been due to Information Gain Stability. It is observed that the Information Gain provided by the attributes remained constant even after the data was binned. The purity of the nodes did not decrease when the change was made from numeric to nominal.

Metric	Before	After	change
Accuracy	100%	100%	-No change
Precision	100%	100%	No change
Recall	100%	100%	No change
Specificity	100%	100%	No change

